LLMBO-MO: Large Language Model-Augmented Multiobjective Bayesian Optimization for Battery Fast-Charging Protocol Design

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***Abstract*—Fast-charging protocol design requires balancing charging time, peak temperature rise, and degradation under an evaluation budget constrained by electrochemical–thermal simulation cost. Bayesian optimization (BO) is suitable for this regime, but its early search is sensitive to a small initial design, while direct use of large-language- model (LLM) predictions would introduce unverified numerical evidence. We propose LLMBO-MO, an asymmetric multiobjective BO framework that uses an LLM only to propose and prioritize search locations. The framework combines a protocol-domain-screened warm start with temporary weight-conditioned regional preferences; the simulator remains the sole objective oracle, and only evaluated protocols train the Gaussian-process surrogate. Under 56 simulator evaluations and five archived seeds, LLMBO-MO obtains a mean hypervolume (HV) of 0.3848 on Chen2020, compared with 0.3778 for ParEGO, and exceeds NSGA-II, DISK, and PIMD by 17.9%–30.8%. On Ecker2015, the observed HV advantage is 4.11% at evaluation 30 and 2.18% at the final budget. A separate ten-seed prompt study shows a mean HV increase of 0.05828 over random initialization after Holm correction ($p = 0.005859$). The component evidence identifies screened warm starting as the more stable contributor, whereas an independent benefit from regional coupling is not established. These results support a restricted-evidence design pattern for injecting language-model priors into simulation-driven engineering optimization without treating generated text as physical evidence.**

***Index Terms*—Battery charging, Bayesian optimization, large language models, lithium-ion batteries, multiobjective optimization.**

I. Introduction

Fast-charging protocol design must reconcile vehicle availability with thermal and degradation limits. Increasing charge current shortens charging time but intensifies heat generation, lithium plating, and solid-electrolyte-interphase growth, creating a trade-off among charging speed, peak temperature rise, and cycle life [1]–[3]. The protocol must also satisfy voltage, temperature, state-of-charge (SOC), and current constraints. Physical cycling tests are slow, and even reduced-order electrochemical–thermal simulations restrict the number of protocols that can be evaluated in one design campaign. The practical problem is therefore to identify a useful Pareto set from a tightly limited evidence budget.

Bayesian optimization (BO) addresses expensive black-box optimization by using a probabilistic surrogate and an acquisition function to allocate evaluations [4]. BO has been applied successfully to closed-loop battery experiments and simulation-based charging design [5]–[7], while multiobjective formulations expose the trade-off surface instead of fixing one preference in advance [8]. Its early iterations nevertheless remain sensitive to the initial design: with only a few observations, uninformative starting points can consume a substantial fraction of the available budget. Electrochemical priors can mitigate this problem [9], but they require calibration data for the relevant cell and operating condition.

Large language models (LLMs) provide a complementary source of initialization and search preferences. Recent work uses LLMs for BO initialization, region preference, candidate sampling, and low-fidelity prediction [10]–[12]. In battery design, however, an LLM proposal is neither a physical measurement nor a feasibility certificate. Treating generated numerical values as observations would transfer evidential authority from the simulator to a model that has not seen the specific cell. The central challenge is therefore to exploit contextual knowledge without allowing language-model output to define objective values or Pareto dominance.

We address this challenge with LLMBO-MO, an asymmetric interface between an LLM and a ParEGO-style multiobjective BO backbone. Before optimization, the LLM proposes a protocol pool from a natural-language description of the battery model, protocol variables, objectives, and design heuristics. Candidates are parsed, repaired to the protocol domain, screened, and selected as a diverse portfolio; random points complete the initial design. The LLM score affects ranking only, and every selected protocol must be simulated before it enters the Gaussian-process (GP) training set. This screened warm start supplies an informative prior over search locations without introducing pseudo-observations.

During an early guidance window, the LLM may also provide weight-conditioned point or region preferences. Validated regions supply additional candidate-pool starts for expected-improvement optimization; malformed, degenerate, or out-of-domain proposals are discarded. The framework also defines an optional bounded posterior-mean coupling for component analysis, but the main Chen2020 archive uses candidate-pool expansion under ordinary EI and accepts no posterior-mean shift. After the guidance window, optimization proceeds without further LLM input. This separation makes the role of the language model explicit: it can suggest where to evaluate, whereas only numerical optimization and simulation determine the evidence.

We evaluate the retained framework under 56 simulator evaluations. Across five archived Chen2020 seeds, LLMBO-MO obtains a mean HV of 0.3848, compared with 0.3778 for ParEGO, and exceeds NSGA-II, DISK, and PIMD by

17.9%–30.8%. On Ecker2015, the observed difference over ParEGO is 4.11% at evaluation 30 and 2.18% at the final budget. A separate ten-seed prompt experiment isolates initialization content and finds an HV increase of 0.05828 over random initialization after Holm correction. The component archive assigns the largest isolated mean change to the screened warm start; it does not establish a systematic additional benefit from regional coupling.

The main contributions are as follows:

1) An asymmetric LLM–BO architecture that separates advisory language-model preferences from simulator-validated objective evidence. The LLM influences candidate generation and ranking but never supplies GP targets or Pareto values.
2) A protocol-domain-screened, diversity-aware warm start together with a temporary weight-conditioned candidate-pool interface. Both mechanisms operate before simulation and preserve the numerical BO backbone as the decision authority.
3) A traceable fixed-budget evaluation on two SPMe parameterizations, including a five-method Chen2020 comparison, an Ecker2015 budget-dependent analysis, a ten-seed prompt study, and a component study that identifies both the supported mechanism and the present evidence boundary.

The remainder of the paper is organized as follows. Section II reviews charging optimization, expensive multiobjective optimization, and LLM-assisted BO. Section III defines the protocol, objectives, constraints, and simulator. Section IV presents LLMBO-MO. Section V reports the numerical evidence and limitations, and Section VI concludes the paper.

II. Related Work

A. Fast-Charging Protocol Optimization

Bayesian optimization (BO) has emerged as the leading sample-efficient paradigm for charging protocol design because it uses a probabilistic surrogate and an acquisition function to allocate a small evaluation budget. Attia *et al.* demonstrated closed-loop BO with physical cells, combining early lifetime prediction with sequential protocol selection [5]. Jiang and co-workers established BO for constrained minimum-time charging, studying acquisition function choice, protocol dimensionality, and operational constraint handling [6], [7]. Wang and Jiang extended this line to the multiobjective setting via Chebyshev decomposition, jointly optimizing charging time, temperature rise, and capacity loss [8]. These studies motivate the three-objective formulation used here and establish BO as a strong baseline for fixed-budget comparison.

Several recent efforts accelerate BO-based charging design through additional numerical structure. Mesh-adaptive search hybridizes BO with direct search grids [13]. Online digital-twin models embed BO inside a continuously updated simulation environment [14]. Parallel satisficing Thompson sampling distributes evaluations across multiple acquisition strategies [15]. Other work incorporates early-life capacity prediction with noisy expected hypervolume improvement [16], a semi-empirical aging-model prior [9], or section-wise electrochemical constraints validated on physical cells [17]. A unifying characteristic of these accelerators is that their guidance derives from numerical models, related-task data, or physical measurements. None exploit the broad but imprecise domain knowledge encoded in large language models (LLMs), nor do they provide a mechanism for incorporating textual advice whose influence is bounded and recoverable. LLMBO-MO addresses this gap: it uses natural-language task descriptions to supply a screened warm-start and bounded regional preferences, while the simulator remains the sole source of evaluated objective values.

B. Expensive Multiobjective Optimization

Multiobjective BO for expensive black-box functions follows two broad strategies. Scalarization-based methods decompose the vector-valued problem into a family of single-objective subproblems. ParEGO uses augmented Tchebycheff scalarization with a single Gaussian process (GP) surrogate and expected improvement (EI), requiring one GP fit per iteration [18]. Constraint-aware extensions model the probability of satisfying expensive inequality constraints alongside the objective [19]. Direct hypervolume-based methods instead optimize a multiobjective acquisition function. Differentiable expected hypervolume improvement (qEHVI) and its noise-aware variant (qNEHVI) enable gradient-based candidate search over the full Pareto frontier [20], [21]. Surrogate-assisted evolutionary algorithms such as DISK and PIMD use Kriging predictions with distribution- or infill-based selection [22], [23]; implementations are available in PlatEMO [24].

LLMBO-MO adopts a ParEGO-style decomposition backbone. This choice ensures that each iteration fits a single scalar GP, matching the per-iteration cost of the baseline and enabling a controlled fixed-budget comparison. The LLM guidance operators are designed to integrate with this backbone without altering its computational scaling: warm-start proposals and regional preferences modify only the initial design and the acquisition surface during an early guidance window, after which the optimizer resumes as ordinary ParEGO.

C. LLM-Assisted Bayesian Optimization

Three recent lines of work integrate LLMs into BO, each assigning a different level of authority to the language-model output.

LLAMBO uses LLMs for zero-shot initialization, surrogate reasoning, and candidate sampling within a standard BO loop [10]. The LLM acts as a general-purpose assistant that proposes design points and interprets optimization progress in natural language. However, LLAMBO treats the LLM as a source of numerical predictions, which introduces fidelity concerns in engineering settings where prediction errors can produce physically infeasible designs.

The preference-guided framework of Yuan *et al.* introduces point and region preferences that shift the surrogate mean while preserving the posterior covariance [11]. This approach is the closest to ours: it uses LLM output to bias the acquisition function rather than to replace simulator evaluations. However, the preference shift is permanent—once applied, the surrogate mean remains altered for all subsequent iterations—and the magnitude of the shift is not explicitly bounded relative to posterior uncertainty.

LABO takes a multi-fidelity perspective, treating inexpensive LLM predictions as low-fidelity observations that inform a discrepancy model with a gating rule that determines when to query the LLM versus the true simulator [12]. While this can reduce evaluation cost, it enters LLM-generated values into the training data, creating a dependence on LLM prediction quality that may be difficult to characterize in constrained battery design.

LLMBO-MO differs from all three in the authority it assigns to language-model output. It does not use LLM predictions as surrogate observations (unlike LLAMBO and LABO) and does not allow LLM preferences to permanently alter the surrogate (unlike the preference-guided framework). Instead, it restricts LLM influence to two bounded, temporary touchpoints—a screened warm-start and an early-window regional guidance operator—each of which discards unusable language-model output before simulator evaluation. This design bounds the authority of the LLM and preserves an unguided numerical fallback, but it does not imply that every LLM-guided optimization trajectory must outperform standard BO.

III. Fast-Charging Optimization Problem

This section defines the protocol parameterization, the three competing objectives, and the physical constraints that together form the constrained multiobjective optimization problem. The coupled electrochemical–thermal simulator that evaluates each candidate protocol is described in Sections III-C and III-D. Figure 1 summarizes the formulation: protocol-feasible coordinates are mapped through the battery simulator to objective values and constraint checks, and a constrained Pareto search updates the protocol proposal.

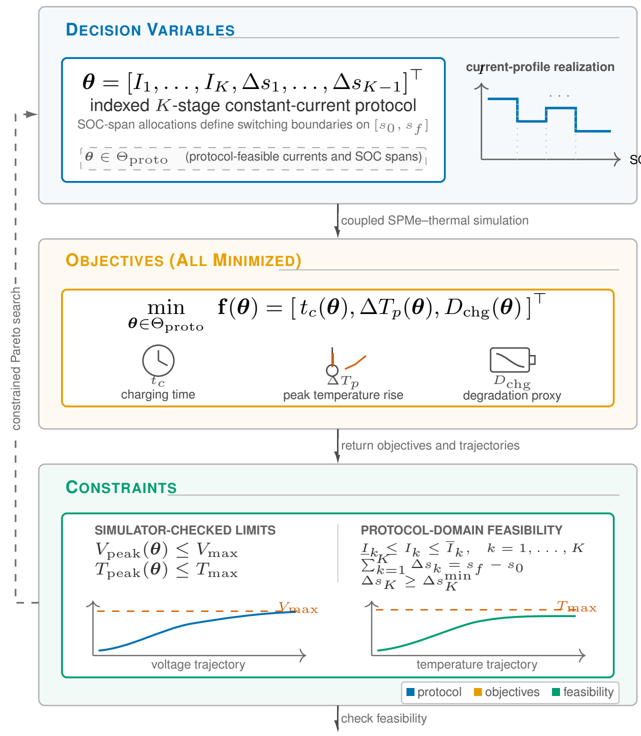

Fig. 1. Fast-charging design problem formulation. Protocol-feasible current and SOC-span coordinates are mapped through the coupled battery evaluator to three minimized objectives and simulator-checked operational limits; constrained Pareto search updates the protocol proposal.

A. SOC-Domain Multistage Protocol Representation

A charging protocol is a sequence of constant-current stages that takes the battery from an initial state of charge s_0 to a target s_f. Following established practice in BO-based charging design [7], [8], each stage is parameterized by its current level and the SOC span it delivers, rather than by a fixed duration. The SOC-domain representation makes switching boundaries explicit: the delivered capacity in each stage depends on the current level, so stage duration is a dependent quantity determined by transferred charge rather than an independent decision variable. This avoids the coupling between current and time that would complicate constraint handling and numerical stability in a time-domain representation.

For a K-stage protocol, let I_k and Δs_k denote the current and SOC span of stage k. The independent decision vector, the dependent terminal span, and the total charging time are

$$\begin{aligned} \boldsymbol{\theta} &= [I_1, \ldots, I_K,\ \Delta s_1, \ldots, \Delta s_{K-1}]^\top, \\ \Delta s_K &= s_f - s_0 - \sum_{k=1}^{K-1} \Delta s_k, \\ t_c(\boldsymbol{\theta}) &= \kappa_t\, Q_{\text{eff}} \sum_{k=1}^{K} \frac{\Delta s_k}{I_k^{\text{cell}}}, \end{aligned} \tag{1}$$

where Q_{eff} is the effective cell capacity, $I_k^{\text{cell}} = \phi_{\text{cell}}(I_k)$ maps a protocol-level current coordinate to the physical cell current via a fixed, cell-specific convention, and κ_t converts charge to time units. The protocol domain Θ_{proto} contains all box-feasible current and SOC-span coordinates satisfying

$\Delta s_K \geq \Delta s_K^{\min}$, which excludes degenerate terminal stages while guaranteeing completion of the prescribed SOC interval. A stricter preferred span Δs_K^{soft} is used only during LLM warm-start candidate ranking (Section IV) and does not enter the simulator constraint set.

This indexed representation accommodates any finite K; the optimization method does not depend on a particular stage count. Following [7], [8], $K = 3$ is used in all experiments.

B. Objectives and Constraints

Three competing quantities are minimized. **Charging time** $t_c(\boldsymbol{\theta})$ is computed analytically from the protocol coordinates via (1). **Peak temperature rise** $\Delta T_p(\boldsymbol{\theta})$ is the maximum cell temperature during the charge minus the initial temperature, obtained from the coupled SPMe–thermal simulation (Section III-C). **Degradation proxy** $D_{\text{chg}}(\boldsymbol{\theta})$ is a control-oriented algebraic severity factor evaluated on trajectory-averaged SOC, temperature, and current; its construction is detailed in Section III-D. Smaller values of D_{chg} indicate less severe average operating conditions under the common proxy, not a prediction of capacity loss in absolute units.

The constrained multiobjective problem is

$$\begin{aligned} \min_{\boldsymbol{\theta}\in\Theta_{\text{proto}}} \quad & \mathbf{f}(\boldsymbol{\theta}) = \begin{bmatrix} t_c(\boldsymbol{\theta}) \\ \Delta T_p(\boldsymbol{\theta}) \\ D_{\text{chg}}(\boldsymbol{\theta}) \end{bmatrix}, \\ \text{s.t.} \quad & V_{\text{peak}}(\boldsymbol{\theta}) \leq V_{\max}, \\ & T_{\text{peak}}(\boldsymbol{\theta}) \leq T_{\max}. \end{aligned} \tag{2}$$

The protocol domain Θ_{proto} enforces current bounds and completion of $[s_0, s_f]$. The simulator checks the path-dependent voltage and temperature limits $V_{\max}$ and $T_{\max}$, which are fixed for each cell parameterization. A protocol that violates either limit or fails to complete the SOC interval is marked infeasible and excluded from the evaluated Pareto set.

The three objectives are inherently conflicting. Higher currents reduce t_c but increase both ΔT_p (through Joule heating) and D_{chg} (through accelerated degradation kinetics at elevated temperature and SOC). The Pareto front therefore represents the set of protocols for which no objective can be improved without worsening another, and the optimization goal is to approximate this front under a tightly limited number of simulator evaluations.

C. Electrochemical–Thermal Simulation Model

Each protocol is evaluated by the same numerical model, so differences among optimizers do not arise from different objective evaluators. The simulator combines a Single Particle Model with electrolyte dynamics (SPMe) and a spatially lumped thermal balance. The SPMe retains solid-phase diffusion, electrolyte transport, and interfacial reaction kinetics at lower cost than a full porous-electrode model [26]; the implementation uses PyBaMM [27].

The standard lumped thermal balance and the reported temperature objective are

$$\begin{aligned} C_{\text{th}} \frac{\mathrm{d}T}{\mathrm{d}t} &= \dot{Q}_{\text{gen}} - hA(T - T_{\text{amb}}), \\ \Delta T_p(\boldsymbol{\theta}) &= \max_t T(t) - T_{\text{init}}, \end{aligned}$$

where C_{th} is the effective thermal capacitance, $\dot{Q}_{\text{gen}}$ collects electrochemical, ohmic, and reversible heat, and hA is the lumped heat-transfer coefficient. Because the model is spatially lumped, ΔT_p measures peak cell-level temperature rise; it does not resolve thermomechanical loading or local temperature gradients.

The Chen2020 parameter set is used for the LG INR21700-M50 case [28]. The Ecker2015 parameter set provides a second cell parameterization [29]. Cell-specific capacity, voltage, and temperature limits are applied, while the optimization logic remains unchanged. Successive protocol stages are simulated sequentially, and the terminal electrochemical and thermal state of one stage initializes the next.

D. Protocol-Level Degradation Proxy

A single protocol simulation does not resolve long-horizon capacity fade. Accordingly, the third objective is an empirical degradation proxy, rather than a prediction of capacity lost during one charge. Let $\bar{s}$, $\bar{T}$, and $\bar{I}$ denote the time averages of SOC, temperature, and current over the completed charge, with $\bar{s}$ represented in the simulator's percentage-SOC convention. The implementation evaluates

$$\begin{aligned} \chi(\bar{s}, \bar{T}, \bar{I}) &= (a_s \bar{s} + b_s) \exp\left(\frac{\beta_I \bar{I} - E_a}{R\bar{T}} \right), \\ D_{\text{chg}}(\boldsymbol{\theta}) &= \kappa_D Q_{\text{eff}} \left(\frac{\chi(\bar{s}, \bar{T}, \bar{I})}{q_a} \right)^{1/\gamma}. \end{aligned} \tag{3}$$

Here a_s, b_s, β_I, E_a, R, q_a, γ, and κ_D are fixed proxy coefficients and are identical for all optimizers; none is re-estimated from the optimization results. This control-oriented algebraic construction is related to the severity-factor family of semi-empirical aging models [25].

The scale factor κ_D preserves the implemented objective scale. We report D_{chg} in arbitrary proxy units (a.u.), not as percentage capacity loss. Smaller values indicate less severe average operating conditions under this common proxy. Because the proxy uses trajectory averages, it does not separately resolve short high-SOC, high-current, or high-temperature excursions; this limitation is considered when interpreting the Pareto set.

Together, the protocol realization and coupled simulator map $\boldsymbol{\theta}$ to $\mathbf{f}(\boldsymbol{\theta})$ and a feasibility flag. Bayesian optimization treats this mapping as an expensive black box and does not access the internal electrochemical states when selecting the next protocol.

IV. LLM-Augmented Multiobjective Bayesian Optimization

A. Design Principle and Overall Workflow

LLMBO-MO treats large-language-model (LLM) output as an advisory preference rather than as an objective observation. This design principle follows from the problem setting: in constrained battery design, an LLM response is neither a physical measurement nor a feasibility guarantee, and it must never enter the evidence chain that determines the Pareto set. The framework accordingly restricts LLM interaction to two controlled, temporary touchpoints: a screened warm-start portfolio constructed before the first BO iteration, and weight-conditioned regional preferences expressed during an early guidance window of T_{L} iterations. Protocol-domain repair, simulator evaluation, surrogate fitting, and acquisition maximization remain purely numerical operations. Unevaluated LLM output never becomes a Gaussian-process (GP) target, and any guidance that fails parsing, repair, or validation gates has zero numerical effect.

Figure 2 illustrates the two advisory paths and their relationship to the common multiobjective BO backbone. At iteration t, a trade-off weight $\boldsymbol{w}^{(t)}$ defines a scalar subproblem. Feasible historical observations are transformed and re-scalarized, the GP is fitted, and expected improvement (EI) selects the next simulator evaluation.

The remainder of this section is organized as follows. Section IV-B describes the screened LLM warm-start. Section IV-C details the objective transformation and ParEGO decomposition backbone shared by LLMBO-MO and the baseline. Section IV-D presents the weight-conditioned region preference interface. Section IV-E derives the bounded posterior-covariance coupling that translates regional preferences into a clipped acquisition bias. Section IV-F summarizes the complete procedure.

B. Protocol-Domain-Screened LLM Warm Start

A central challenge in BO is the cold start: before task-specific observations accumulate, the surrogate is shaped primarily by the initial design and the GP prior. Random initialization, while unbiased, can waste a substantial fraction of the evaluation budget on unpromising regions when the design space is large and constrained. Model-based priors can mitigate this [9] but require per-cell, per-condition calibration. LLMBO-MO instead uses the LLM to propose an initial candidate set from a natural-language description of the task. Because the LLM output is not assumed to be physically accurate, every candidate passes through a protocol-domain screening pipeline before evaluation.

The warm-start prompt supplies the cell-model context, variable bounds, protocol geometry (K-stage constant-current, SOC-domain representation), and battery-inspired scheduling heuristics such as decreasing current profiles and avoiding extreme terminal spans. The LLM returns numerical candidate protocols, each accompanied by a self-reported confidence score. The repair-and-screen pipeline processes the raw LLM output: non-numerical entries are discarded, duplicate and non-finite values are removed, box-violating coordinates are clipped, and candidates that cannot satisfy the dependent terminal SOC span $\Delta s_K \geq \Delta s_K^{\min}$ are rejected. These checks establish protocol-domain validity; operational voltage and temperature feasibility is determined only by simulation.

Let $h_{\mathrm{ws}}(\boldsymbol{\theta}) \in [0,1]$ denote the LLM-reported candidate score, which is used only for ranking. A soft penalty discourages a terminal SOC span below the preferred boundary,

$$p_{\mathrm{soft}}(\boldsymbol{\theta}) = \mathrm{clip}\left(\frac{\Delta s_K^{\mathrm{soft}} - \Delta s_K}{w_{\mathrm{soft}}}, 0, 1\right).$$

Let $\mathbb{I}_{\mathrm{mono}}(\boldsymbol{\theta})$ indicate a nonincreasing current sequence. The selected portfolio is built greedily using

$$\begin{aligned} J_{\mathrm{ws}}(\boldsymbol{\theta}) = h_{\mathrm{ws}}(\boldsymbol{\theta}) - \omega_p p_{\mathrm{soft}}(\boldsymbol{\theta}) + \omega_m \mathbb{I}_{\mathrm{mono}}(\boldsymbol{\theta}) \\ + \omega_d d_{\min}(\boldsymbol{\theta}, \mathcal{C}_{\mathrm{sel}}), \end{aligned} \tag{4}$$

where normalized decision coordinates are denoted by a hat and

$$d_{\min}(\boldsymbol{\theta}, \mathcal{C}_{\mathrm{sel}}) = \begin{cases} 0, & \mathcal{C}_{\mathrm{sel}} = \varnothing, \\ \min\limits_{\mathbf{c} \in \mathcal{C}_{\mathrm{sel}}} \|\widehat{\boldsymbol{\theta}} - \widehat{\mathbf{c}}\|_2, & \text{otherwise.} \end{cases}$$

The first selected candidate therefore receives no artificial diversity bonus. Random protocol-domain-valid points complete the initial design to maintain exploration coverage, and only simulator-evaluated candidates are appended to the optimization database $\mathcal{D}$.

This warm-start design has two properties that distinguish it from both random initialization and model-prior methods. First, it requires no per-cell calibration: the same natural-language prompt template is used across parameterizations, with only the variable bounds and cell context updated. Second, protocol-domain screening guarantees that every LLM-sourced initial point satisfies the same box and SOC-completion constraints as the random points, ensuring a fair initialization comparison.

C. Objective Transformation and ParEGO Backbone

LLMBO-MO shares its scalarization and surrogate backbone with ParEGO [18] to enable a controlled fixed-budget comparison. The three objectives span different numerical ranges (charging time in seconds, temperature rise in Kelvin, degradation proxy in arbitrary units). A logarithmic transform is applied to charging time and the degradation proxy to compress their dynamic range; temperature rise remains on its physical scale:

$$\begin{aligned} \tilde{\mathbf{f}}(\boldsymbol{\theta}) &= [\ell_t(\boldsymbol{\theta}),\, \Delta T_p(\boldsymbol{\theta}),\, \ell_D(\boldsymbol{\theta})]^\top, \\ \ell_t(\boldsymbol{\theta}) &= \log_{10}\max(t_c/t_{\mathrm{ref}}, \varepsilon_t), \\ \ell_D(\boldsymbol{\theta}) &= \log_{10}\max(D_{\mathrm{chg}}/D_{\mathrm{ref}}, \varepsilon_D). \end{aligned}$$

The same transformation is applied to LLMBO-MO and ParEGO, so differences in hypervolume are not artifacts of different objective scalings. For objective i, let $R_{i,t}^{\mathrm{hist}}$ be the current observed range and define the global transformed range by $R_i^{\mathrm{glob}} = \tilde{y}_i^{\mathrm{ref}} - \tilde{y}_i^{\mathrm{ideal}}$, where the ideal and reference coordinates are mapped through the same objective transformation. The scale used by the implementation is

$$s_{i,t} = \begin{cases} R_i^{\mathrm{glob}}, & R_{i,t}^{\mathrm{hist}} < \rho_R R_i^{\mathrm{glob}}, \\ R_{i,t}^{\mathrm{hist}}, & \text{otherwise.} \end{cases}$$

This hybrid scaling uses the full global range when the early observed range is too narrow, preventing premature exploitation of an incorrectly estimated scale.

Following ParEGO [18], the three objectives ($m = 3$) are combined through an augmented Tchebycheff target,

$$\begin{aligned} \bar{f}_{i,t}(\boldsymbol{\theta}) &= \frac{|\tilde{f}_i(\boldsymbol{\theta}) - \tilde{z}_{i,t}^*|}{s_{i,t}}, \\ g_t(\boldsymbol{\theta} \mid \boldsymbol{w}) &= \max_i w_i \bar{f}_{i,t}(\boldsymbol{\theta}) + \rho_{\mathrm{aug}} \sum_{i=1}^{m} w_i \bar{f}_{i,t}(\boldsymbol{\theta}), \end{aligned}$$

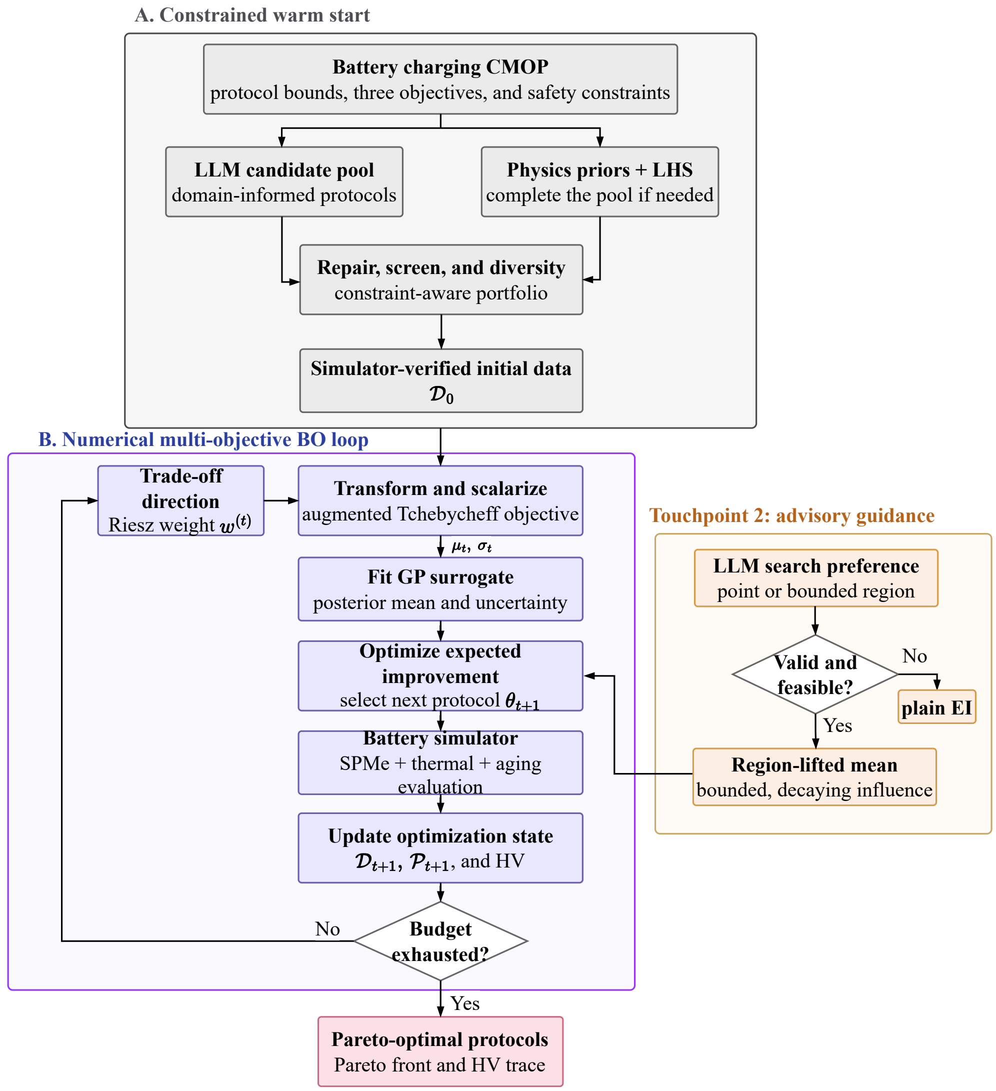

Fig. 2. Unified LLMBO-MO workflow. *A:* domain-informed LLM candidates and numerical initialization points are repaired against the protocol domain and evaluated by the simulator. *B:* the numerical multiobjective BO loop performs scalarization, GP fitting, EI optimization, and simulator evaluation. The LLM enters only through the two advisory touchpoints; rejected guidance recovers plain EI, and only simulator outputs update the optimization state.

with $\boldsymbol{w} \geq \mathbf{0}$ and $\mathbf{1}^\top \boldsymbol{w} = 1$. The weight set is initialized on a Das–Dennis simplex lattice and dispersed by projected Riesz-energy relaxation [30]; weights are visited without replacement before reshuffling.

Standardized scalar targets are modeled by an ARD Matérn-$5/2$ GP with a white-noise term on normalized decision variables. Kernel parameters are estimated by marginal likelihood with multiple restarts. The fitted posterior mean μ_t, standard deviation σ_t, and covariance are the only statistical quantities used by the acquisition function.

D. Weight-Conditioned LLM Region Preference

The second LLM touchpoint operates during an early guidance window of T_{L} iterations. After the initial portfolio has been evaluated and the GP surrogate has accumulated some task-specific evidence, the LLM is queried with richer context to elicit weight-conditioned search preferences.

The prompt at iteration $t \leq T_{\mathrm{L}}$ includes the current scalarization weight $\boldsymbol{w}^{(t)}$, the per-objective observed ranges and current best values, a selection of recently evaluated protocols with their objective values, and a summary of optimization progress. The LLM response schema permits three outcomes: a raw-coordinate point, a hyperrectangular region defined by lower and upper bounds, or an explicit declaration of no preference. Providing a schema with a no-preference option is deliberate: when the LLM lacks useful insight for a particular weight configuration, silence is preferable to an unfounded suggestion.

A received point or region is processed through the same repair-and-screen pipeline as the warm-start candidates. Coordinates are parsed, clipped to variable bounds, repaired to Θ_{proto}, and validated. A point preference is converted to a local hyperrectangle by adding a fixed-width margin in each coordinate dimension. All accepted proposals are then subjected to width and volume checks: a region whose span in any dimension falls below a minimum width, or whose normalized volume exceeds a maximum threshold, is rejected.

These checks prevent degenerate (single-point) and vacuous (full-space) regions from entering the acquisition stage.

Anchors are drawn uniformly from each accepted box and serve two independent roles. First, they act as external restarts for numerical EI maximization, providing alternative starting points to the standard multi-start L-BFGS-B procedure. Second, they may enter the bounded posterior-covariance coupling described in Section IV-E. These roles are separable: accepting a region does not force a GP-mean adjustment, and disabling the coupling leaves the restart-augmented EI search well defined.

The guidance window closes after T_{L} iterations. Beyond this point, the LLM is no longer queried, and the optimizer continues as ordinary ParEGO. This temporal bound prevents stale or increasingly irrelevant LLM suggestions from influencing later convergence stages, where the surrogate is already well-informed by simulator data.

E. Bounded Posterior-Covariance Preference Coupling

When an accepted region passes all validation checks, its anchors may be used to construct a bounded adjustment to the acquisition mean. The design objective is to make regions preferred by the LLM marginally more attractive to EI while explicitly limiting the adjustment in standardized scalar-target units. The coupling proceeds in three stages: anchor scoring, covariance projection, and gated attenuation.

Anchor scoring. Let $\mathcal{G}_t = \{\mathbf{g}_{j,t}\}_{j=1}^{M}$ be the accepted anchors drawn from validated regions. Each anchor is scored by a weighted combination of its EI value, posterior uncertainty, and novelty relative to the evaluated design. With $z(\cdot)$ denoting a clipped across-anchor standardization,

$$\begin{aligned} n_{j,t} &= \min_{\mathbf{x}\in\mathcal{D}_t} \|\widehat{\mathbf{g}}_{j,t} - \widehat{\mathbf{x}}\|_2, \\ s_{j,t} &= \eta_{\mathrm{EI}} z(\log\max\{\mathrm{EI}_{j,t}, \varepsilon_{\mathrm{EI}}\}) + \eta_\sigma z(\sigma_{j,t}) + \eta_n z(n_{j,t}), \\ v_{j,t} &= \frac{\exp(s_{j,t}/T_v)}{\sum_{\ell=1}^{M}\exp(s_{\ell,t}/T_v)}. \end{aligned} \tag{5}$$

If anchor scoring is numerically unavailable, the implementation uses uniform weights $v_{j,t} = 1/M$. Thus $\mathbf{v}_t \geq \mathbf{0}$ and $\mathbf{1}^\top \mathbf{v}_t = 1$.

Covariance projection. Let $\mathbf{\Sigma}_{GG,t}$ be the posterior anchor covariance and $\mathbf{\Sigma}_{\theta G,t}$ the candidate-to-anchor covariance. With covariance jitter ε_K and variance floor ε_v,

$$\begin{aligned} V_{G,t} &= \mathbf{v}_t^\top(\mathbf{\Sigma}_{GG,t} + \varepsilon_K \mathbf{I})\mathbf{v}_t, \\ \rho_t(\boldsymbol{\theta}) &= \frac{\mathbf{\Sigma}_{\theta G,t}\mathbf{v}_t}{\sigma_t(\boldsymbol{\theta})\sqrt{\max\{V_{G,t}, \varepsilon_v\}}}. \end{aligned}$$

The scalar $\rho_t(\boldsymbol{\theta}) \in [-1, 1]$ is the posterior correlation between the candidate point and the weighted anchor portfolio. A positive value indicates that the candidate lies in a region positively correlated with the LLM-preferred anchors; only positive correlations are used for the mean adjustment.

Gated attenuation. Three independent factors gate the adjustment magnitude. The region-quality gate combines anchor concentration and score dispersion:

$$\begin{aligned} H_t &= -\sum_{j=1}^{M} v_{j,t}\log(v_{j,t} + \varepsilon_H), \\ \bar{H}_t &= \frac{H_t}{\max\{\log M, \varepsilon_H\}}, \\ q_t &= \mathrm{clip}\big[\eta_H(1-\bar{H}_t) + (1-\eta_H)\tanh\!\big(\mathrm{std}(\mathbf{s}_t)\big),\, 0,\, 1\big]. \end{aligned}$$

The LLM-reported confidence $c_t = \mathrm{clip}(h_t^{\mathrm{LLM}}, 0, 1)$ and a bounded trust factor $\tau_t \in [0, 1]$ complete the gate product $r_t = c_t \tau_t q_t$. The evaluated main mode holds this factor fixed and skips posterior-mean coupling; adaptive trust updates are an optional implementation mode rather than a property assumed by the principal results. The early-window attenuation $a_t = \mathrm{clip}(1 - (t-1)/T_{\mathrm{L}}, 0, 1)$ linearly ramps the influence to zero by the end of the guidance window. The acquisition-time adjustment is then

$$\begin{aligned} \delta_t(\boldsymbol{\theta}) &= \mathrm{clip}(a_t \lambda_{\max} r_t [\rho_t(\boldsymbol{\theta})]_+, 0, \delta_{\max}), \\ \mu_t^{\mathrm{L}}(\boldsymbol{\theta}) &= \mu_t(\boldsymbol{\theta}) - \delta_t(\boldsymbol{\theta}), \qquad (\sigma_t^{\mathrm{L}})^2(\boldsymbol{\theta}) = \sigma_t^2(\boldsymbol{\theta}). \end{aligned} \tag{6}$$

Because the scalar Tchebycheff target is minimized, reducing the GP mean near positively correlated anchors makes those regions more attractive to EI. The explicit clipping guarantees $0 \leq \delta_t(\boldsymbol{\theta}) \leq \delta_{\max}$ in standardized target units, so the adjustment magnitude is explicitly controlled. The operation preserves three critical invariants: it does not alter fitted GP targets, it does not modify posterior uncertainty, and it does not create pseudo-observations. If any gate is zero ($q_t = 0$, $c_t = 0$, or $\tau_t = 0$), then $r_t = 0$ and the posterior mean used by EI is unchanged. Accepted regional restarts may still expand the numerical candidate pool, so this equality is a coupling fallback rather than a guarantee that the complete optimization trajectory matches unguided BO. Table I lists the core numerical settings that instantiate these operators.

F. Complete Optimization Procedure

Let μ_t^{A} equal μ_t^{L} when every coupling check passes and μ_t otherwise. Standard minimization EI is evaluated using this active mean. The candidate pool combines the incumbent, Gaussian and uniform samples, multi-start L-BFGS-B solutions, and accepted regional restarts. Candidates are repaired to Θ_{proto} before simulation; path-dependent feasibility is known only after evaluation.

V. Numerical Validation and Discussion

A. Evaluation Setup

All quantitative evidence in this paper is simulation based. The SPMe formulation in Section III-C is evaluated with the Chen2020 and Ecker2015 parameter sets [28], [29]. Each protocol is scored by charging time t_c, peak temperature rise ΔT_p, and the degradation proxy D_{chg} (Section III-D). BO-based runs use six initialization evaluations followed by 50 optimization iterations, for a total of 56 simulator calls.

Hypervolume (HV) is the evaluation metric; higher is better. All methods within one battery parameterization use the same transformed objectives and a fixed benchmark-specific reporting box. The reporting box is common to the

Algorithm 1 LLMBO-MO

Require: Simulator $\mathcal{S}$, LLM $\mathcal{L}$, total budget N, initial budget n_0, weights $\mathcal{W}$, guidance horizon T_{L}
Ensure: Evaluated nondominated set $\mathcal{P}$
Query $\mathcal{L}$ for a warm-start pool; repair and rank by (4)
Add selected LLM proposals and random protocol-domain-valid points
Evaluate n_0 initial protocols and initialize $\mathcal{D}$
for $t = 1$ to $N - n_0$ **do**
  Select $\boldsymbol{w}^{(t)}$; transform and re-scalarize feasible data
  Fit the GP to $g_t(\cdot \mid \boldsymbol{w}^{(t)})$
  if $t \leq T_{\mathrm{L}}$ **then**
    Query $\mathcal{L}$ for a point/region and validate its anchors
    Add accepted anchors as restarts; attempt (6)
  end if
  Maximize EI using μ_t^{A} and select $\boldsymbol{\theta}_{t+1}$
  Evaluate $\mathcal{S}(\boldsymbol{\theta}_{t+1})$ and append the result to $\mathcal{D}$
  Update $\mathcal{P}$ and the GP training subset using feasible results
end for
return $\mathcal{P}$

TABLE I
FIXED-BUDGET PROTOCOL AND ARCHIVED LLMBO-MO CONFIGURATION

Setting	Archived value
Battery parameterizations	Chen2020; Ecker2015
Initialization / BO calls	6 / 50
Total simulator calls	56
LLMBO-MO initialization	3 screened LLM + 3 random
ParEGO initialization	6 random
Chen2020 archived seeds	8409–8413
Ecker2015 archived seeds	8409–8413
Guidance horizon / anchors	16 / 64
Region influence in main Chen runs	early candidate-pool expansion
Posterior-mean coupling in main Chen runs	disabled
LLM aliases	archive dependent; recorded per run

methods being compared but differs between Chen2020 and Ecker2015, so absolute HV values are interpreted only within one parameterization. Reported uncertainty is one sample standard deviation.

The main tables summarize five archived seeds per method. For the Chen2020 LLMBO-MO and ParEGO comparison, seed 8409 is taken from the archive underlying Fig. 3, while seeds 8410–8413 come from the retained comparison archives. The runs share the parameterization, method preset, 56-call budget, and HV definition, but were not executed in one synchronized batch. They are therefore reported as descriptive five-seed evidence; no inferential or equivalence claim is made. The archive manifest records the source and configuration of every entry.

ParEGO is the primary BO comparator because it shares the scalarization–GP backbone with LLMBO-MO [18]. The Chen2020 archive additionally contains NSGA-II and the surrogate-assisted evolutionary algorithms DISK and PIMD [22], [23], [31] under the same 56-evaluation cap. Matched qEHVI and qNEHVI runs are not present in the archive; their absence is treated as an evaluation limitation rather than as evidence against those methods [20], [21].

The LLM aliases vary across retained archives, so cross-parameterization differences cannot be attributed to battery transfer alone. Within every run, the simulator remains the sole objective oracle and only simulator-evaluated protocols enter the GP training set and Pareto archive.

TABLE II
CHEN2020 FINAL HV OVER FIVE ARCHIVED SEEDS AND 56 EVALUATIONS

Method	Final HV ↑	LLMBO-MO advantage
LLMBO-MO	$\mathbf{0.3848 \pm 0.0073}$	–
ParEGO	0.3778 ± 0.0169	1.85%
NSGA-II	0.3265 ± 0.0220	17.9%
DISK	0.3085 ± 0.0267	24.8%
PIMD	0.2941 ± 0.0195	30.8%

TABLE III
ECKER2015 HV AT INTERMEDIATE AND FINAL BUDGETS

Method	Evaluation 30	Evaluation 56
ParEGO	0.4870	0.5540 ± 0.0121
LLMBO-MO	**0.5070**	$\mathbf{0.5660 \pm 0.0104}$

TABLE IV
CHEN2020 WARM-START PROMPT STUDY OVER TEN SEEDS

Initialization source	Final HV ↑
Random	0.31062 ± 0.04303
Detailed generic prompt	0.31371 ± 0.04675
Experimental battery prompt	$\mathbf{0.36890 \pm 0.00824}$

B. Chen2020 Five-Method Comparison

Table II shows that LLMBO-MO obtains the highest observed mean HV in the retained Chen2020 set. Its mean is 1.85% above ParEGO and 17.9%–30.8% above the evolutionary or transfer baselines. The larger dispersion of ParEGO reflects substantial run-to-run variability, including the seed-8409 trajectory visualized below. Because the five entries combine compatible historical archives rather than a single synchronized campaign, these differences characterize the retained evidence and are not used for a significance claim.

The representative seed-8409 case in Fig. 3 ends at 0.3848 for LLMBO-MO and 0.3523 for ParEGO, a descriptive difference of 9.2%. This case illustrates how the screened initialization and early candidate-pool expansion can place the search in a favorable part of the protocol space. It is not substituted for the five-seed statistics in Table II.

C. Ecker2015 Budget-Dependent Comparison

Figure 4 exposes a budget-dependent effect. At evaluation 30, LLMBO-MO reaches an HV of 0.5070, compared with 0.4870 for ParEGO, corresponding to a 4.11% intermediate- budget advantage. At the full 56-evaluation budget, the respective means are 0.5660 and 0.5540, leaving a smaller but positive 2.18% final difference. The result supports faster progress in the middle of the search and a retained final advantage; it does not establish a parameterization-independent effect.

D. Warm-Start Prompt Study

The strongest controlled evidence for the language component is a separate Chen2020 prompt study using ten seeds and a shorter 26-evaluation budget. All variants share the downstream BO procedure and differ only in the initial-candidate source.

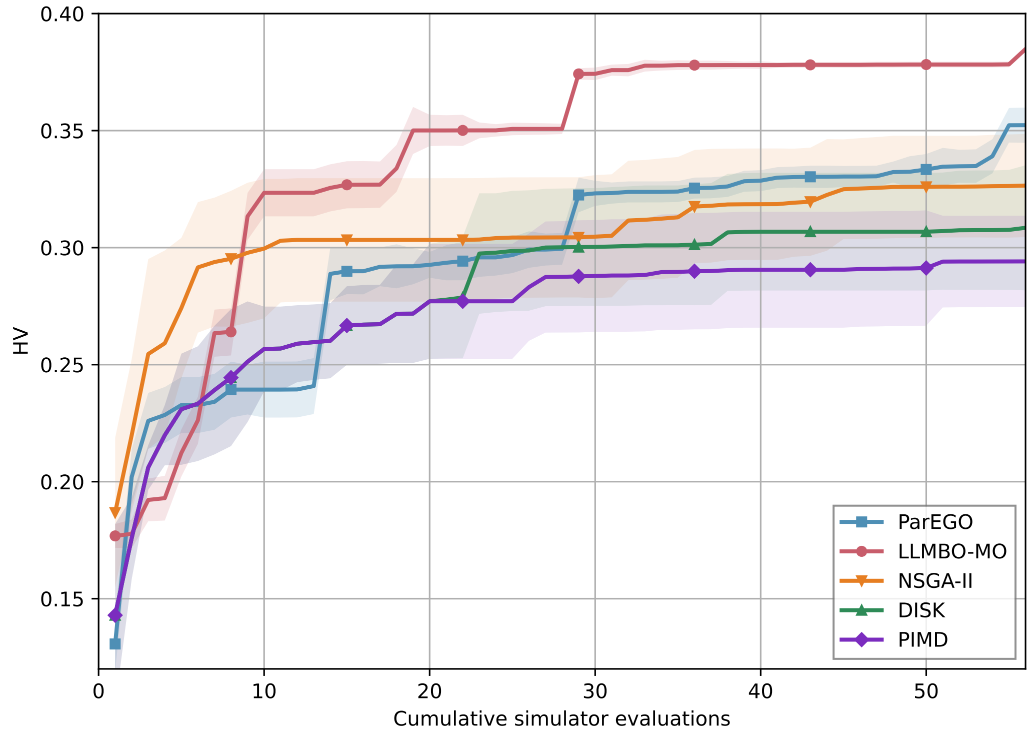

Fig. 3. Chen2020 five-method search progress under the 56-evaluation cap. The LLMBO-MO and ParEGO center lines are representative seed-8409 trajectories; their shaded proxy envelopes are derived from five auxiliary archived traces. The NSGA-II, DISK, and PIMD lines are five-run means and their bands show one standard deviation. The formal five-seed final statistics are reported in Table II.

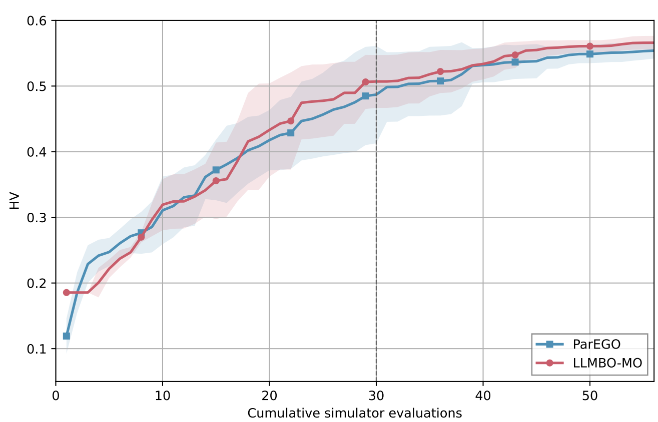

Fig. 4. Ecker2015 HV over five archived seeds. Lines are five-seed means and bands show one sample standard deviation. The dashed line marks evaluation 30, where the intermediate-budget comparison is reported.

The experimental battery prompt improves mean HV over random initialization by 0.05828. The paired contrast remains nonzero after Holm correction ($p = 0.005859$) and has standardized effect size $d_z = 1.309$. This result supports the value of battery-specific prompt content for short-budget initialization; it does not isolate regional guidance or extend directly to the 56-evaluation main comparison.

E. Component Study

The Chen2020 component archive separates the screened warm start from regional guidance under the 56-evaluation cap.

TABLE V
CHEN2020 COMPONENT STUDY OVER FIVE ARCHIVED SEEDS

Variant	Final HV ↑	Difference from plain BO
Plain BO	0.3836 ± 0.0113	–
Warm start	**0.3902 ± 0.0087**	+0.0066
Regional guidance	0.3862 ± 0.0125	+0.0026
Complete LLMBO-MO	0.3900 ± 0.0067	+0.0064

The values in Table V are descriptive because the retained regional variants contain an external-restart difference and the historical archive does not preserve a complete source hash.

The screened warm start provides the largest isolated mean change and reduces dispersion relative to plain BO. Regional guidance alone has a smaller positive mean difference, while the complete configuration is essentially level with warm start alone. The component evidence therefore supports warm starting as the more stable contributor but does not support an additive or systematic regional benefit.

F. Pareto Protocol and Trajectory Interpretation

Figure 5 links the objective-space envelope to time-domain behavior. The marked protocols expose the expected conflict among charging time, temperature rise, and the degradation proxy; no marked point minimizes all three objectives. The selected LLMBO-MO schedule reaches the target SOC earlier than the two illustrative alternatives at the cost of a larger temperature excursion. These trajectories are simulated and are not measurements of lifetime or protection-system performance.

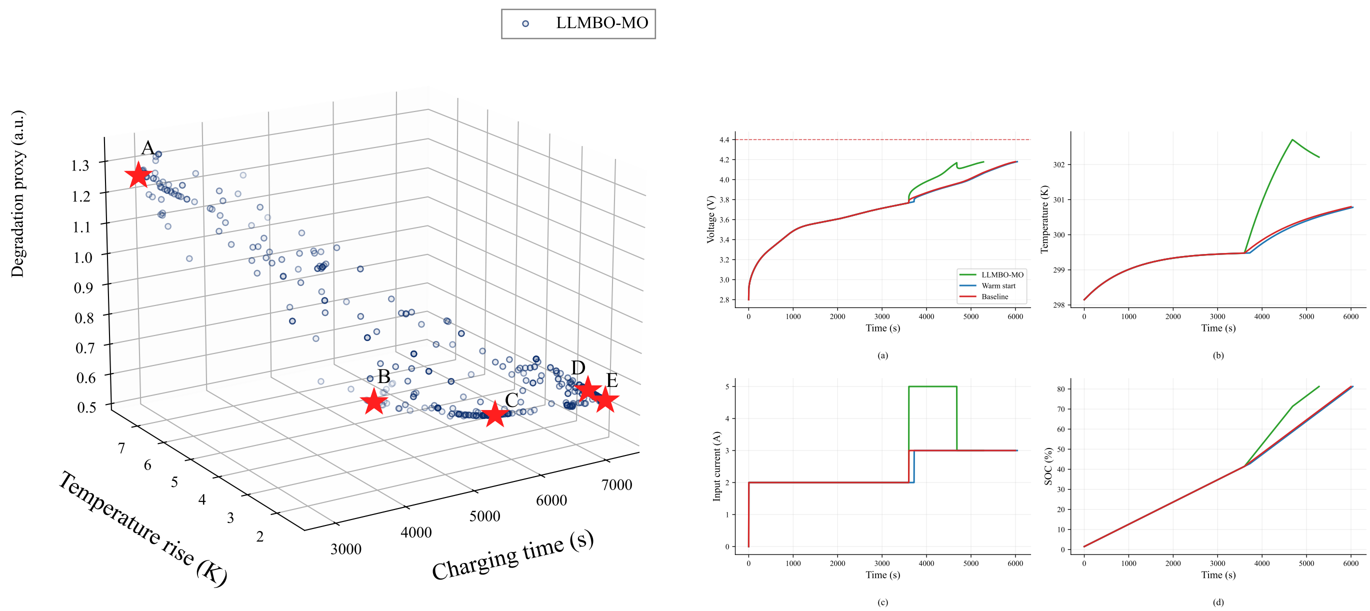

Fig. 5. Chen2020 protocol interpretation. *Left:* evaluated LLMBO-MO protocols in the three-objective space, pooled across five archived runs for visualization; A–E mark representative trade-offs. *Right:* simulated current, voltage, temperature, and SOC trajectories for selected LLMBO-MO, warm-start, and ParEGO protocols.

G. Limitations

The evidence has five principal limitations. First, all quantitative results are simulation based, and D_{chg} is an uncalibrated trajectory-averaged proxy rather than a measured capacity-fade endpoint. Second, the Chen2020 five-seed summary combines compatible historical archives rather than one synchronized execution batch. Third, matched qEHVI and qNEHVI results are unavailable, so the present study does not establish state-of-the-art MOBO performance. Fourth, the LLM backend differs across archives, preventing attribution of cross-parameterization differences to battery transfer. Finally, the component archive contains an acquisition-restart confound and does not establish an independent benefit from regional coupling.

VI. Conclusion

This paper introduced LLMBO-MO for simulation-limited multiobjective battery fast-charging design. Its central idea is to separate advisory language-model knowledge from evidence-grade numerical evaluation: the LLM proposes and prioritizes search locations, while the simulator remains the sole objective oracle and only evaluated protocols train the GP surrogate. The resulting framework combines a protocol-domain-screened warm start with temporary weight-conditioned candidate-pool guidance and an optional bounded posterior-mean coupling.

Across five archived Chen2020 seeds, LLMBO-MO achieves a mean HV of 0.3848, compared with 0.3778 for ParEGO, and exceeds the three retained evolutionary or transfer baselines by 17.9%–30.8%. On Ecker2015, the observed advantage is 4.11% at evaluation 30 and 2.18% at the final budget. The ten-seed prompt study provides the strongest controlled language-component evidence, with an HV increase of 0.05828 over random initialization after multiplicity correction. The component study indicates that screened warm starting is the more stable contributor; the retained data do not establish an independent systematic benefit from regional coupling.

The current conclusions are limited to SPMe simulation, two parameterizations, an uncalibrated degradation proxy, and partially mixed historical archives. Matched qEHVI/qNEHVI comparisons, synchronized multi-seed runs under one code snapshot, controlled LLM-backend studies, and physical-cell replay are needed before claiming state-of-the-art performance, battery transfer, or measured lifetime improvement. These extensions will determine whether the bounded-advisory interface can retain its early-search benefit under stronger numerical baselines and experimentally validated degradation endpoints.

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